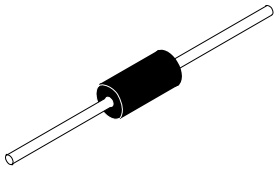
# Small Signal Diode

# 1N5282



**AXIAL LEAD (DO−35)**
**CASE 017AG**
(Color Band Denotes Cathode)

## ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted)

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $V_{RRM}$ | Maximum Repetitive Reverse Voltage | 80 | V |
| $I_{F(AV)}$ | Average Rectified Forward Current | 200 | mA |
| $I_{FSM}$ | Non−repetitive Peak Forward Surge Current<br>Pulse Width = 1.0 s<br>Pulse Width = 1.0 μs | <br>1.0<br>4.0 | <br>A<br>A |
| $T_{STG}$ | Storage Temperature Range | −65 to +200 | °C |
| $T_J$ | Operating Junction Temperature | 175 | °C |

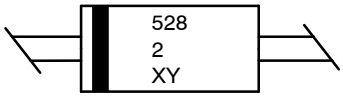
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. These ratings are based on a maximum junction temperature of 200°C.
2. These are steady state limits. The factory should be consulted on applications involving pulsed or low duty cycle operations.

## THERMAL CHARACTERISTICS

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $P_D$ | Power Dissipation | 500 | mV |
| $R_{\theta JA}$ | Thermal Resistance, Junction to Ambient | 300 | °C/W |

## MARKING DIAGRAM




5282 = Specific Device Code
XY = Date Code

## ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| 1N5282 | AXIAL LEAD (Pb−Free, Halide Free) | 5000 Units / Bulk |
| 1N5282TR | | 10000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

## ELECTRICAL CHARACTERISTICS ($T_A$ = 25°C unless otherwise noted)

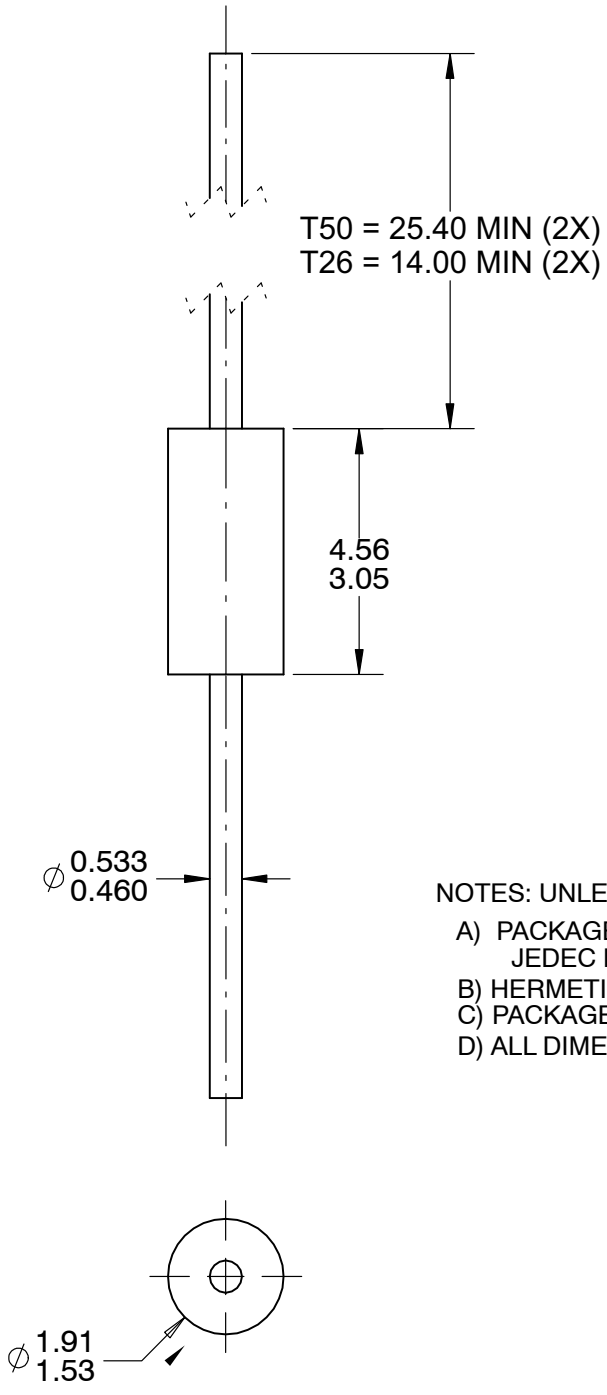
| Symbol | Parameter | Test Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $V_R$ | Breakdown Voltage | $I_R$ = 5 μA | 80 | – | – | V |
| $V_F$ | Forward Voltage | $I_F$ = 0.1 mA<br>$I_F$ = 1.0 mA<br>$I_F$ = 10 mA<br>$I_F$ = 100 mA<br>$I_F$ = 300 mA<br>$I_F$ = 500 mA | 0.45<br>0.55<br>0.67<br>0.80<br>0.92<br>1.05 | –<br>–<br>–<br>–<br>–<br>– | 0.49<br>0.60<br>0.725<br>0.90<br>1.1<br>1.3 | V |
| $I_R$ | Reverse Current | $V_R$ = 55 V<br>$V_R$ = 55 V, $T_A$ = 150°C | – | – | 100<br>100 | nA<br>μA |
| $C_T$ | Total Capacitance | $V_R$ = 0, f = 1.0 MHz | – | – | 2.5 | pF |
| $t_{rr1}$ | Reverse Recovery Time | $I_F$ = $I_R$ = 10 mA, $R_L$ = 100 Ω<br>Irr = 1.0 mA | – | – | 4 | ns |
| $t_{rr2}$ | Reverse Recovery Time | $I_F$ = $I_R$ = 200 mA, $R_L$ = 100 Ω<br>Irr = 20 mA | – | – | 4 | ns |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

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**AXIAL LEAD**
CASE 017AG
ISSUE O

DATE 31 AUG 2016




NOTES: UNLESS OTHERWISE SPECIFIED

A) PACKAGE STANDARD REFERENCE:
JEDEC DO−204, VARIATION AH.
B) HERMETICALLY SEALED GLASS PACKAGE.
C) PACKAGE WEIGHT IS 0.137 GRAM.
D) ALL DIMENSIONS ARE IN MILLIMETERS.

| | | | |
|---|---|---|---|
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| **DESCRIPTION:** | **AXIAL LEAD** | | **PAGE 1 OF 1** |

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